

The second semi-meander diagonal: a cut proof

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Statement and attribution. Label $2n$ vertices by $0, \dots, 2n-1$, and fix the lower rainbow $x \leftrightarrow 2n-1-x$. An upper matching is a noncrossing perfect matching. Count it only when its union with the rainbow is one loop. Let $T(n, k)$ count these matchings with $2(k-1) + (n \bmod 2)$ upper arches crossing the midpoint, and put $D_j(n) = T(n, \lfloor n/2 \rfloor - j + 1)$. Thus $D_2(n)$ has $n-4$ midpoint-crossing arches. We prove, for every integer $n \geq 4$,

$$D_2(n) = \frac{n^2 + 2n + (n \bmod 2) - 20}{2}. \quad (1)$$

This is the conjecture in Hunter Hogan's OEIS A400429, revision 16. The same formula is credited to the resummed prediction of Di Francesco, Golinelli and Guitter (1996), Appendix D. The proof below is independent; no priority claim is made.

1. Encoding the cut. Use labels $0, \dots, n-1$ in each half, reversing the right half so the two ends of each rainbow arch have the same label. Exactly four vertices in each half belong to local upper pairs. A vertex strictly between the ends of a local arch cannot cross the midpoint: that would cross the arch. Consequently a local arch is either an adjacent pair, or an arch on four consecutive vertices with its adjacent inner pair. Each half is therefore specified uniquely by $p = (a, b)$ with

$$1 \leq a < b < n, \quad b \geq 3,$$

whose local pairs are

$$\{a-1, a\}, \quad \{s, b\}, \quad s = \begin{cases} a-2 & b = a+1, \\ b-1 & b \geq a+2. \end{cases} \quad (2)$$

In the first case the pairs are nested; in the second they are disjoint. Let U_p be the remaining labels. Given p, q for the two halves, noncrossing forces the r th smallest label of U_p to join the r th smallest label of U_q . This also constructs a noncrossing matching for every such pair: no unmatched vertex lies inside a local arch. Write $R : U_p \rightarrow U_q$ for this increasing bijection.

Contract the lower rainbow. The resulting undirected graph $G(p, q)$ has vertices $0, \dots, n-1$, the four local pairs from (2), and edges $\{x, R(x)\}$ for $x \in U_p$. Contraction preserves connectivity, so $D_2(n)$ is exactly the number of ordered pairs (p, q) for which $G(p, q)$ is connected.

2. Connectivity classification. If 0 is unmatched in both halves, $R(0) = 0$ and 0 is isolated from all other vertices. The same holds for $n-1$. Hence one prefix has either $a = 1$ or $(a, b) = (2, 3)$, and one prefix has $b = n-1$. The complete list of connected pairs is the following three families and their exchanged copies:

$$\begin{array}{lll} A & p = (1, b), \ q = (c, n-1) & 3 \leq b \leq n-2, \ b-1 \leq c \leq n-2, \ c \not\equiv b \pmod{2}; \\ B & p = (1, n-1), \ q = (c, c+1) & 2 \leq c \leq n-2; \\ C & p = (2, 3), \ q = (c, n-1) & 3 \leq c \leq n-2, \ c \text{ odd}. \end{array} \quad (3)$$

Here are explicit reductions proving both directions of this list.

Family B. Suppose $p = (1, n-1)$. If $q = (c, d)$ with $d \geq c+2$, the set $\{0, \dots, c\}$ is closed under every edge. Indeed each half has two paired labels in that set, so their unmatched lists have the same initial segment length and the rank bijection stays inside it. The local pairs also stay inside or outside. This is a proper nonempty set. For $n = 4$, the only such pair is $p = q = (1, 3)$; its two components are $\{0, 1\}$ and $\{2, 3\}$. If instead $d = c+1$, necessarily $c \geq 2$, and direct deletion of the four paired labels gives

$$R(x) = \begin{cases} x-2 & x < c, \\ x+2 & x \geq c \end{cases} \quad (x \in U_p = \{2, \dots, n-3\}).$$

Repeated steps reach 0 or 1 below c , and $n-2$ or $n-1$ above c . The local pairs of p join each of these two endpoint groups; the local pair $\{c-2, c+1\}$ of q joins the groups. Thus the graph is connected.

Family A. Suppose $p = (1, b)$, $3 \leq b \leq n-2$, and $q = (c, n-1)$. If $c \leq b-2$, the same initial-segment argument makes $\{0, \dots, c\}$ a proper closed set. Now assume $c \geq b-1$. On U_p the rank map is

$$R(x) = \begin{cases} x-2 & x < b, \\ x-2 & x > c+2 \text{ and } c+2 \leq n-1, \\ x-4 & \text{otherwise.} \end{cases} \quad (4)$$

These values are nonnegative and strictly smaller than x . Repeatedly apply (4) until reaching a root in $\{0, 1, b-1, b\}$, and denote that root by $t(x)$. The local pairs of p join the root groups $L = \{0, 1\}$ and $H = \{b-1, b\}$ separately. All rank edges preserve t . For completeness an explicit formula for the root is

$$\begin{aligned} t(x) &= x \bmod 2 && \text{if } x \leq b-2; \\ y &= \begin{cases} c+1 + ((x-c-1) \bmod 2) & x > c+2, \ c+2 \leq n-1, \\ x & \text{otherwise,} \end{cases} && (5) \\ r &= (y+3-b) \bmod 4, \quad t(x) = \begin{cases} (b-3+r) \bmod 2 & r < 2, \\ b-3+r & r \geq 2 \end{cases} && \text{if } x \geq b-1. \end{aligned}$$

Formula (5) fixes the four roots and satisfies $t(R(x)) = t(x)$; induction along the strictly decreasing steps proves it. Substitution of the two pairs (2) of q into (5) gives

	$\{c-1, c\}$	$\{s, n-1\}$
$c \equiv b \pmod{2}$	same root group	same root group
$c \not\equiv b \pmod{2}$	opposite root groups	not needed

The substitution uses $s = c-2$ when $c = n-2$, and $s = n-2$ otherwise. Thus equal parity preserves two components; opposite parity supplies a bridge joining every vertex. This proves precisely the condition in A.

Family C. Suppose $p = (2, 3)$ and $q = (c, n-1)$. When $c = 1$ this is exchanged B. Otherwise $c \geq 2$, $U_p = \{4, \dots, n-1\}$, and

$$R(x) = \begin{cases} x-4 & x \leq c+2, \\ x-2 & x > c+2. \end{cases}$$

All vertices reduce to roots 0, 1, 2, 3. Their terminal labels are

$$t(x) = \begin{cases} x \bmod 4 & x \leq c+2, \\ (c+1) \bmod 4 & x > c+2, \ x - (c+3) \text{ even,} \\ (c+2) \bmod 4 & x > c+2, \ x - (c+3) \text{ odd.} \end{cases} \quad (6)$$

The pairs of p join $\{0, 3\}$ and $\{1, 2\}$. If c is even, substituting the pairs of q into (6) preserves these groups. If c is odd, $\{c-1, c\}$ projects to $\{0, 1\}$ or $\{2, 3\}$, joining them. This proves C. The endpoint restrictions preceding (3) reduce every pair to A, B, C or an exchange. The case $(2, 3)$ with $b = n-1$ forces $n = 4$ and has already been covered. Inspection of (3) shows all six families are disjoint.

3. Counting. Set $m = n - 4$. In A put $r = n - 1 - b$, so $1 \leq r \leq m$. The allowed first endpoint of q is exactly $c = b - 1 + 2k$, $0 \leq k \leq \lfloor r/2 \rfloor$. B is parametrized by $c = 2, \dots, n - 2$, and C by $c = 3, 5, \dots \leq n - 2$. Their cardinalities are therefore

$$|A| = \sum_{r=1}^m (1 + \lfloor r/2 \rfloor) = m + \lfloor m^2/4 \rfloor, \quad |B| = m + 1, \quad |C| = \lfloor (m + 1)/2 \rfloor.$$

For the sum identity, $m = 0, 1$ give $0, 1$; increasing m by two adds $(1 + \lfloor (m + 1)/2 \rfloor) + (1 + \lfloor (m + 2)/2 \rfloor) = m + 3$, which is also the increment of $m + \lfloor m^2/4 \rfloor$. Since exchanging the two halves is a bijection onto the other three disjoint families,

$$D_2(n) = 2 \lfloor m + \lfloor m^2/4 \rfloor + (m + 1) + \lfloor (m + 1)/2 \rfloor \rfloor. \quad (7)$$

If $m = 2h$, (7) equals $2h^2 + 10h + 2$; if $m = 2h + 1$, it equals $2h^2 + 12h + 8$. Substituting $n = m + 4$ gives (1) in each case. At $n = 4$, A and C are empty, and B and its exchange contribute one each: $D_2(4) = 2$. The two prefix pairs are $((1, 3), (2, 3))$ and its exchange. $\square$

4. What the Lean development proves. The precise source is repository revision 5c0f5b1b3f3b07e72bf5ba6cc8b7daf4b2cc428d. `Model.lean` defines upper matchings as fixed-point-free involutions on $\text{Fin}(2n)$, imposes noncrossing, counts midpoint-crossing arches, and defines one-loop connectivity by reachability through upper and rainbow steps. Stages 1–3 construct the cut encoding and its inverse and transfer connectivity. Stages 3–4 prove the endpoint obstructions and terminal connectivity tests. Stage 5 proves the disjoint classification (3), the finite-family bijections and their cardinalities. The final theorem `result` states

$$\text{Nat.card}\{M : \text{UpperMatching}(n) \mid M.\text{winding} = n - 4, M.\text{oneLoop}\} = (n^2 + 2n + n \% 2 - 20)/2 \quad (n \geq 4).$$

Here the formal arithmetic uses natural-number division. The identification with the OEIS objects is given by the arch definition and the midpoint identity above. This note reports the existing formalization; it does not report a fresh Lean validation run. The D_3 conjecture is outside the statement of this development.

5. A concrete next statement. OEIS A400429 also conjectures, for every $n \geq 6$,

$$D_3(n) = \frac{1}{288} [13n^4 + 24n^3 - 284n^2 - 4560n + 17920 + (n \bmod 2)(18n^2 - 72n + 189) - 64\mathbf{1}_{3|n}].$$

This diagonal has $n - 6$ crossing arches, hence six locally paired vertices in each half. A next project would enumerate these local types independently for $6 \leq n \leq 30$, share counts by type and residue class modulo 6, and formalize the six-endpoint cut bijection. Comparison should use direct enumeration rather than a table generated from the conjectured formula. These are proposed contributions, not completed computations or a proof of D_3 .

Sources.

- Hunter Hogan, OEIS A400429, revision 16: <https://oeis.org/A400429>.
- P. Di Francesco, O. Golinelli and E. Guitter, *Meanders: A Direct Enumeration Approach* (1996), Appendix D, (D.9)–(D.14): <https://arxiv.org/abs/hep-th/9607039>.
- Lean theorem and imported proof modules: [SemiMeanderSecondDiagonal.lean](#) at the revision above.